Representative values are for the US 0.1–10 Bq/m$^3$, UK and Germany ~3, and 10 (average) for France. The concentration in air above ground depends also on temperature and wind conditions.

Many ores contain small amounts of uranium. During processing, uranium and/or its daughters may enter the product, causing a radioactive contamination problem. For example, when apatite is used to produce phosphoric acid, the gypsum by-product contains all the radium originally present, producing a γ-ray and inhalation hazard from Rn-daughters, making it unsuitable for building material.

Radon concentrations in indoor air may be quite high, depending on site and building material. The $^{226}$Ra content e.g. in German building materials varies from >500 (gypsum) to 60 (brick) Bq/kg; slag used in Poland contains <800, Italian tuff ≈280, concrete in Hungary ≈13, and white bricks in the UK only ≈4 Bq/kg. The indoor concentration of radon also depends on the way in which the house is built and used (poorly ventilated, etc). In the US it varies between <1 to >1000 Bq/m$^3$. Authorities no longer recommend very tight houses, as suggested in the 1970's to reduce heating costs. At levels <70 Bq/m$^3$ the Rn-hazard is considered negligible. In Sweden (pop. 8 M people) 50% of the houses have 70–200 Bq/m$^3$, while 40 000 houses have been classified as "Rn-houses", i.e. their indoor concentration of Rn plus daughters exceed 400 Bq/m$^3$. If the ground is the main source of radon, ventilation of the basement may be sufficient to declassify a "Rn-house".

Even coal contains small amounts of uranium, 4 to 300 kBq/ton, a typical value being 20 kBq/ton. When coal is burnt, the more volatile U-daughter products are released into the atmosphere, while the less volatile ones are deposited in the dust filters. A 1 GW$_e$ coal-fired power plant typically releases 60 GBq $^{222}$Rn and 5 GBq $^{210}$Pb+Po annually, while ~3 MBq daughter products are obtained per ton of fly ash.

## 13.7. DISEQUILIBRIUM

Uranium ores at surface level are usually identified from the penetrating γ-ray emissions of the daughter products. However, it has been observed that some peat, which absorb uranium from local drainage, contain very little of the daughter products because they have been formed so recently that radioactive equilibrium has not been established.

The decay products of uranium passes over 10 elements (Fig. 1.2), all with very different chemical properties. These elements are transported by groundwater (*migrate*), the solute composition of which varies with the surrounding rock/soil minerals. The different elements migrate at different rates due to their different chemistries, dissolving in some areas and precipitating in others. If the mother and daughter in a radioactive chain migrate at different velocities during a time which is short in relation to the daughter half-life, the radioactive equilibrium is disturbed; this is referred to as *disequilibrium*. Such disequilibrium can be used to measure the age of the sample.

Let us consider the essential steps of the uranium decay series:

$$\begin{array}{ccccccccccc} & \alpha+2\beta & & \alpha & & \alpha & & 4\alpha+2\beta & & 2\beta+\alpha & \\ {}^{238}\mathrm{U} & \dashrightarrow & {}^{234}\mathrm{U} & \dashrightarrow & {}^{230}\mathrm{Th} & \dashrightarrow & {}^{226}\mathrm{Ra} & \dashrightarrow & {}^{210}\mathrm{Pb} & \dashrightarrow & {}^{206}\mathrm{Pb} \\ & 4.5\times10^{9}\ \mathrm{y} & & 2.5\times10^{5}\ \mathrm{y} & & 7.5\times10^{4}\ \mathrm{y} & & 1.6\times10^{3}\ \mathrm{y} & & 22\ \mathrm{y} & \\ & |\Leftarrow \approx 10^{6}\ \mathrm{y} & \Rightarrow|\Leftarrow & \approx 3\times10^{5}\ \mathrm{y} & \Rightarrow|\Leftarrow & \approx 10^{4}\ \mathrm{y} & \Rightarrow|\Leftarrow & \approx 100\ \mathrm{y} & \Rightarrow|\Leftarrow & \approx 100\ \mathrm{y} & \Rightarrow| \end{array} \tag{13.4}$$

In this decay series the nuclide pairs which are suitable for determination of ages are associated with the time periods given between the double arrows of the two connected isotopes. For example, the $^{238}$U-decay to $^{234}$U passes over the short-lived intermediates $^{234}$Th ($t_{½}$ 24.1 d) and $^{234}$Pa ($t_{½}$ 1.17 m). The Th-isotope is long-lived enough to follow its own chemistry in a dynamic system. In strongly acidic solutions it forms $Th^{4+}$ ions, while U forms $UO_2^{2+}$ ions; the behavior of these two ions with regard to complex formation (e.g. by carbonates, hydroxyl or humic acid) and solubility is drastically different in neutral waters, leading to different migration rates for the two elements. For example, $^{238}$U may migrate away, while $^{234}$Th is precipitated or sorbed. As a result, when $^{234}$Th decays (via the rapid equilibrium with $^{234}$Pa) to $^{234}$U, the latter is free from $^{238}$U. From the deviation from the original activity ratio of $^{238}U/^{234}U = 1.0$, ($^{238}U$ refers to the concentration of that isotope, see §3.3.2) the time since $^{238}$U and $^{234}$Th separated (i.e. the age of the sample) is deduced.

An example of such a system is the sedimentary uranium deposit in Tono, Japan, studied by Nohara et al. The ground water moves through the area with a velocity of 0.001 to 1 m/y. The specific activity ratio, measured by α- and βγ-spectrometry, of $^{234}U/^{238}U$ is plotted against $^{230}Th/^{234}U$ (a), and $^{226}Ra/^{230}Th$ against $^{210}Pb/^{226}Ra$ (b) for a large number of rock samples in Figure 13.5. If radioactive equilibrium existed all ratios

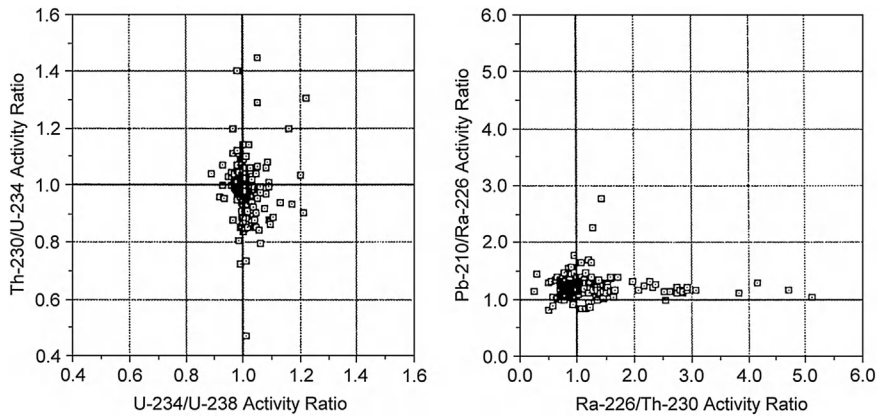



**Figure 13.5** Activity ratios $^{234}U/^{238}U$ vs $^{230}Th/^{234}U$, and $^{226}Ra/^{230}Th$ vs $^{210}Pb/^{226}Ra$ for rock samples from the Tono uranium deposit. *(From Nohara et al.).*

should be 1.0. The observation of deviations from 0.5 to 1.5 in (a) and up to 5 in (b) indicate that U, Th, Ra and Pb have migrated at different velocities in recent times. A detailed analysis yields the age of the U deposit and the migration rates of the daughter elements: the U and Th has not migrated during the past several hundred thousand years (this is probably also the age of the U deposit); Ra has migrated a few meters in the last 10 000 y.

## 13.8. AGE DETERMINATION FROM RADIOACTIVE DECAY

Prior to the discovery of radioactivity, geologists could obtain only poor estimates of the time scale of the evolution of the earth. The oldest geologic materials were assumed to be some 10 million years old, and it was believed that this represented the age of the earth. However, with the discovery of radioactivity early in this century, geologists developed more objective methods for such age determination ("nuclear clocks"). In 1907 B. B. Boltwood obtained a value of $2.2 \times 10^9$ y for the age uranium and thorium minerals, assuming that all U and Th ultimately decayed to lead. Considering how few isotopes in the chains had been discovered at the time, the calculation was surprisingly good. Nuclear clocks have provided primary data on the age and evolution of the earth (*nuclear geo-chronology*) as well as the formation of elements and of the universe (*cosmo-chronology*).

The cosmogenic radionuclides with relatively short half-lives can be used to date materials of more recent origin; e.g., $^{3}H$ for water movements in the geosphere, and $^{14}C$ for organic material of archaeologic interest. Practically all of the primordial radionuclides (Table 13.3) can be used for dating geologic materials: $^{40}K/^{40}Ar$ for igneous (plutonic) rocks (i.e. rocks which have solidified from a rather homogenous melt); $^{87}Rb/^{87}Sr$ for metamorphic and sedimentary rocks; $^{147}Sm/^{143}Nd$ for rock-forming silicate, phosphate and carbonate minerals; $^{187}Re/^{187}Os$ for sulfides and metallic material like iron meteorites, etc.

### 13.8.1. Dating by the $^{14}C$ method

It is reasonable to assume that the production of $^{14}C$ in the atmosphere has been rather constant for at least a million years, which means that equilibrium exists between the rates of formation and decay of the $^{14}C$ in the atmosphere. Moreover, the half-life of $^{14}C$ is sufficient to allow equilibrium between the $^{14}C$ in the atmosphere, the oceans (including precipitations to ocean bottoms), and exchangeable carbon in natural materials. Thus from measurement of the specific radioactivity of carbon, it should be possible to determine when the sample became isolated from its natural environmental compartment.

The discovery that all living organic material has a certain specific radioactivity due to $^{14}C$ led W. Libby to a new method for determination of the age of biological material. This method, which has been of great importance in dating archeological, geological,